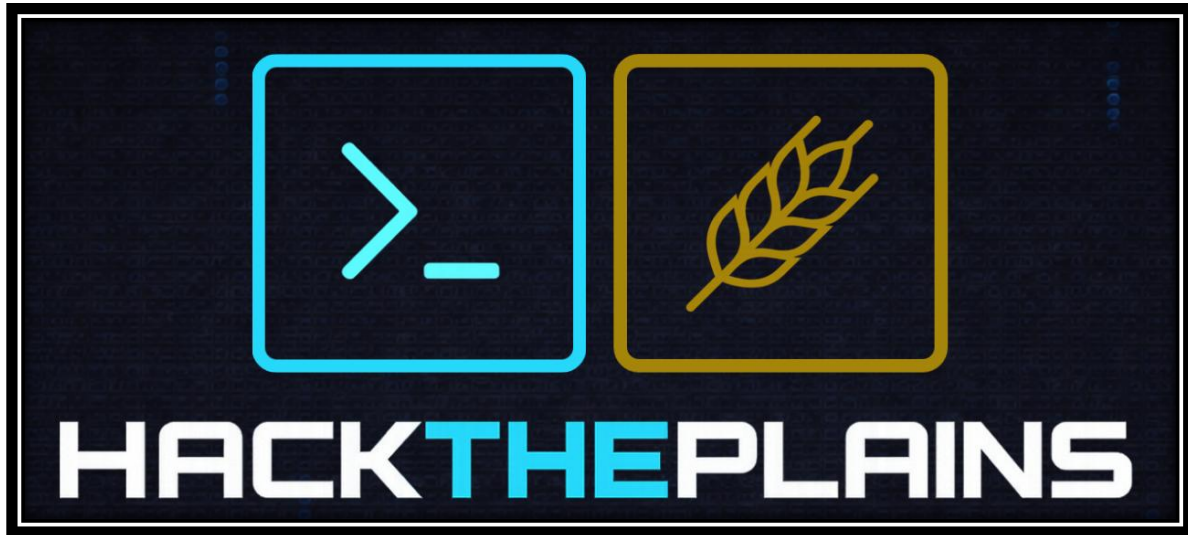




Digital Banking & Channel Optimization

Problem Statement, Dataset Guide & Bonus Challenge
April 10-12, 2026

For Participants

1. Background

Before we get into the challenge, let's set the scene. If you're not from a banking or financial services background, this section will give you the context you need to understand what we're working with and why it matters.

The Shift to Digital Banking

Over the past decade, the way people interact with their banks has changed dramatically. What used to require a trip to a physical branch, like depositing a check, transferring money, or paying a bill, can now be done from a phone or laptop in seconds. Most banks today offer mobile apps and websites where customers can manage nearly everything about their accounts without ever speaking to a person.

This shift has been great for convenience, but it has also created new problems. When something goes wrong in a digital experience, like a transaction that fails, a confusing error message, or a feature that doesn't load, customers get stuck. And when they get stuck digitally, many of them pick up the phone and call the bank's support center, or they drive to a branch for help.

Here's the thing: a successful digital transaction costs the bank a fraction of what a phone call or branch visit costs. When a customer calls because something broke online, the bank is essentially paying twice: once for the failed digital attempt, and again for the human support to fix it. Multiply that across thousands of customers a day, and it adds up fast.

The Balancing Act

Banks want to move more customers to digital channels because it's more efficient and often more convenient. But they can't just shut down phone lines and branches, because not every task works well digitally yet, and not every customer is comfortable with technology. The real challenge is figuring out where digital is falling short, fixing those pain points, and doing it in a way that actually makes the experience better, not just cheaper.

That's exactly what you'll be doing in this hackathon.

2. The Challenge

This year's challenge asks you to step into the shoes of a bank's data team and answer a question that real financial institutions wrestle with every day:

How can a bank improve digital engagement, reduce customer friction, and lower operational cost, without negatively impacting customer experience?

What Makes This Challenge Compelling

- **It's realistic and broadly applicable.** Every bank with digital channels faces these trade-offs.
- **It reflects actual business tensions.** Efficiency gains can't come at the expense of customer satisfaction.
- **It's open-ended.** There is no single correct answer. Judges are looking for thoughtful analysis and strong reasoning, not a pre-determined solution.

What You're Expected to Do

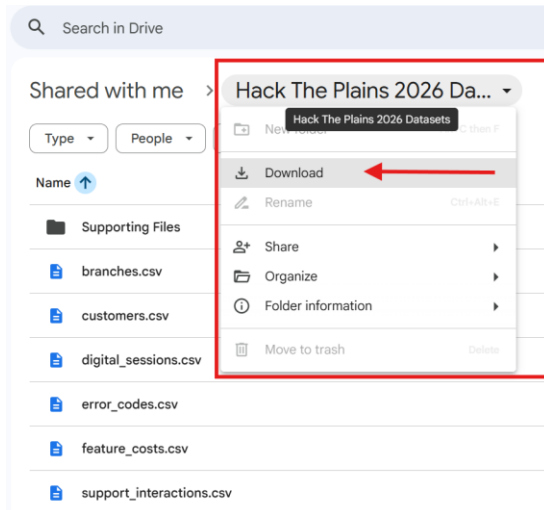
1. **Identify where customers struggle** in digital channels (errors, abandonment, friction).
2. **Understand why customers still call or visit branches.** Is it by choice or because digital failed them?
3. **Connect behavior across multiple systems.** Sessions, support interactions, customer profiles, and costs.
4. **Translate findings into actionable business recommendations** with a clear prioritization framework.

3. Dataset Overview

You will receive a synthetic dataset designed to simulate a real bank's multi-system data environment. Take time to explore each file and understand how they relate to one another.

Link to Google Drive with data: [Data Folder](#)

Download the folder directly on your computer.



Please do not edit or delete any files in the folder.

Files Provided

File	Rows	Description
customers.csv	~355,000	Customer demographics, tenure, segment, products, digital enrollment date, home branch, churn flag
digital_sessions.csv	~2,000,000	Session logs: channel, device, feature used, start/end times, outcome, error code
support_interactions.csv	~650,000	Support contact logs: interaction type, reason code, timestamp, resolution status
branches.csv	40	Branch reference data: code, city, state
feature_costs.csv	11	Average cost per successful and failed session by digital feature
error_codes.csv	10	Error code descriptions and reference

A Note on Data Quality

Like real-world data, this dataset is not perfectly clean. Part of the challenge is discovering what issues exist and deciding how to handle them. Document your decisions as you go, as this will be part of how your work is evaluated.

Note: This dataset is entirely synthetic. It does not represent real customer data from any financial institution. Please do not imply otherwise in your presentation.

4. Guidelines & Deliverables

Event Schedule

The hackathon runs from Friday, April 10 through Sunday, April 12, 2026. Plan your time carefully, especially on Sunday when the deadline is firm.

Friday, April 10

- **6:00 PM** Doors Open & Check-in @ **WSU, Hubbard Hall, Room 231**
- **6:30 PM** Opening Ceremony & Dataset Reveal
- **7:00 PM** Hacking Begins

Saturday, April 11

Teams have the full day to work on their projects. Jimmy (Admin) will be available for assistance throughout the day. Need help? Schedule a meeting virtually or stop by in person at WSU.

Sunday, April 12

- **10:00 AM** Project Submissions Due
- **10:00 AM – 1:00 PM** Judging Period
- **By 3:00 PM** Winners Announced

The winning team will be notified via email and announced on hacktheplains.com. The prize will be presented to the winning team in cash.

Deliverables

Each team must submit the following to their assigned Google Drive folder (a unique link will be provided to each team):

1. **A recorded presentation** (10-15 minutes). Walk the judges through your findings, analysis, and recommendations as if you were presenting to a room of bank executives. Use a slide deck to support your narrative. This is your primary deliverable and carries the most weight.

All team members are expected to participate in the recorded presentation. The easiest way to do this is with Zoom or Google Meet:

- One team member creates and starts a meeting and invites the rest of the team.
 - One team member shares their screen to display the slide deck.
 - The meeting host hits Record before the presentation begins.
 - Team members can take turns speaking through different sections, just as you would in a live presentation.
 - When finished, end the meeting and export as an MP4 file.
 - Save your file as an MP4 and name it clearly, for example: TeamName_Presentation.mp4.
2. **Supporting code and analysis.** Upload your notebooks, scripts, queries, or any other working files that show how you arrived at your conclusions. These don't need to be polished, but they should be organized enough for a judge to follow your process.
 3. **A primary contact phone number.** Include a text file or note with a phone number where the admin or judging team can reach your team for any follow-up questions.
 4. **A team photo** (optional but appreciated!). We'd love to feature your team on the website and social media when announcing results.

All files must be uploaded to your team's Google Drive folder by Sunday, April 12 at 10:00 AM. Late submissions will not be accepted. If you have trouble accessing your folder, reach out to admin@hacktheplains.com.

How You'll Be Scored

Your submission will be scored out of 100 points across four categories:

Category	Points
Data Wrangling & Preparation	30
Analytical Depth & Insight Quality	30
Business Recommendations	20
Presentation & Communication	20

Tips for Success

- **Start with the data.** Don't rush to conclusions. Invest time exploring the data and building a clean foundation.
- **Connect the dots across systems.** The strongest analyses will link digital sessions to support interactions to customer profiles to cost data.
- **Think like a business leader.** Judges want to see recommendations they could act on Monday, not just interesting charts.
- **Document your decisions.** When you make judgment calls about data quality or analysis approach, briefly note why. This shows analytical maturity.
- **Practice your presentation.** Communication is worth 20% of your score. Tell a clear story that a non-technical executive would follow.

Tools & Technology

You may use any tools, languages, or platforms your team is comfortable with. Python, R, SQL, Excel, Tableau, Power BI, all are welcome. There is no restriction on tools, only on the data (use only what's provided).

5. Bonus Challenge: Share Your Story

BONUS POINTS OPPORTUNITY

Want to earn extra points for your team? After the hackathon, at least one team member can write a LinkedIn post that highlights:

- **The problem statement** you tackled and why it matters.
- **Your team's approach and key findings.** What did you discover? What was surprising?
- **Your recommendations** and the impact they could have.

Post Requirements

- **The post should effectively communicate** your solution to a general audience, not just data professionals.
- **Tag all team members** in the post so they get credit.
- **Tag #HackThePlains2026** and include a link to hacktheplains.com.
- **Keep it professional but engaging.** Show your personality and enthusiasm for the work you did.

This is a great opportunity to showcase your data skills to your professional network.

6. Glossary

Not everyone comes from a banking or data background, and that's perfectly fine. Here are some terms you might encounter during the challenge:

Term	Definition
Churn	When a customer closes their account or stops using the bank's services. A "churn flag" in the data indicates whether a customer has left.
Digital Channel	Any non-physical way customers interact with the bank, such as a mobile app or website (as opposed to a branch or phone call).
Session	A single visit or interaction a customer has with a digital channel. It starts when they log in or open the app, and ends when they leave or their session times out.
Session Outcome	The result of a digital session: success (task completed), error (something went wrong), or abandon (the customer left without finishing).
Error Code	A short identifier (like E001, E002) that indicates what type of problem occurred during a session.
Feature	A specific capability within the bank's digital platform, such as bill pay, mobile deposit, balance check, or funds transfer.
Branch	A physical bank location where customers can walk in for service.
Tenure	How long a customer has been with the bank, typically measured in months.
Segment	A category used to group customers by type, such as Retail (everyday consumers) or Business.
Support Interaction	Any time a customer contacts the bank for help, whether by phone, chat, or in person at a branch.
Reason Code	A label that categorizes why a customer contacted support (e.g., ACCOUNT_OPENING, FRAUD_ALERT, PASSWORD_RESET).
Resolution Status	Whether a support interaction was resolved (the customer's issue was fixed) or unresolved.
Orphan Record	A record in one dataset that references an ID (like a customer ID) that doesn't exist in the related dataset.

De-duplication	The process of identifying and removing duplicate records from a dataset.
Cost per Session	How much it costs the bank, on average, for a customer to use a particular digital feature. Failed sessions typically cost more than successful ones.
Digital Enrollment	When a customer signs up to use the bank's online or mobile services.
Friction	Anything that makes it harder, slower, or more frustrating for a customer to complete a task digitally.

Good luck, have fun, and we can't wait to see what you build!